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ORIGINAL PAPER

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Active vision in insects: an analysis of object-directed zig-zag flights in wasps (*Odynerus spinipes*, Eumenidae)

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Abstract Ground-nesting wasps (*Odynerus spinipes*, Eumenidae) perform characteristic zig-zag flight manoeuvres when they encounter a novel object in the vicinity of their nests. We analysed flight parameters and flight control mechanisms and reconstructed the optical flow fields which the wasps generate by these flight manoeuvres. During zig-zag flights, the wasps move sideways and turn to keep the object in their frontal visual field. Their turning speed is controlled by the relative motion between object and background. We find that the wasps adjust their rotational and translational speed in such a way as to produce a specific vortex field of image motion that is centred on the novel object. As a result, differential image motion and changes in the direction of motion vectors are maximal in the vicinity and at the edges of the object. Zig-zag flights thus seem to be a 'depth from motion' procedure for the extraction of object-related depth information.

Key words *Odynerus spinipes* · Wasps · Optical flow · Motion parallax · Active vision

Introduction

Image motion plays an important role in the visual orientation of insects (see Wehner 1981 for a review). When animals move, they generate optical flow in their visual field that is determined by both the parameters of their own movements and by the three-dimensional layout of the environment. From this pattern of image motion, the nervous system could in principle extract egomotion parameters and cues to the 3D structure of the local environment (Koenderink 1986). In insects, the rotational components of egomotion are detected by a set of wide-field, movement-sensitive, directionally selective interneurons which receive input from an array of local motion detectors (reviewed by Hausen 1993). Many of these neurons seem to be involved in flight control, they detect deviations from an intended course and their output mediates appropriate compensatory head or body movements. The most basic reflex in this context is the optomotor response to pattern rotation around the principal rotational axes of an animal.

While during normal flight the animals' own movements and disturbances caused by wind produce image motion by default, some insects perform characteristic scanning movements that seem to be specifically designed to generate motion parallax as a cue to depth. The peering movements of locusts and mantids are a case in point: before jumping, they judge the distance to a landing site by facing it and swaying their bodies from side to side. The faster the image of the landing site moves in the visual field during these scanning movements, the closer it is perceived by the peering insects, as can be judged from their subsequent jump distance (Wallace 1959; Collett 1978; Sobel 1990; Collett and Paterson 1991). Barn owls move in a rather similar way when they inspect a new room or novel objects (Wagner 1989). The learning flights bees and wasps perform when they depart from their nest or from a newly discovered food source also belong to this class of information-processing behaviours. The insects acquire a visual representation of the goal environment in the course of learning flights and their subsequent use of depth cues when pinpointing the goal indicates that the processing of distance information is one specific function of these flights (reviewed by Zeil et al. 1996).

Specially tailored movements are not necessarily a prerequisite for exploiting motion parallax cues. Without apparent scanning movements, honeybees and flies can judge the distance between objects and background

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by motion discontinuities. They prefer to land on edges which provide motion parallax cues and use image speed in a centering response that allows them to fly through holes or along the centre of a tunnel (Lehrer and Srinivasan 1993; review by Srinivasan 1993; Kimmerle et al. 1996). In certain tasks, however, bees have to learn to adopt a flight behaviour that helps them to generate the image motion cues relevant to the task. When bees are asked to detect raised edges between striped surfaces, they learn to suppress their normal tendency to fly parallel to the contours of a striped pattern. Instead, they spend most of their time flying more or less perpendicular to the pattern orientation during their search for the target edges and in this way scan for the motion discontinuities between foreground and background patterns that define these edges (Lehrer and Srinivasan 1994).

In this paper we describe and analyse another class of scanning movements, namely the 'zig-zag flights' that some insects perform in front of objects. We found that these flights are most easily studied in females of a particular species of ground-nesting wasps which respond to novel objects in the vicinity of their nests by vigorously zig-zagging in front of them. We are particularly interested in the consequences of these characteristic flight manoeuvres for visual information processing and therefore reconstructed the optical flow field the insects experience during their zig-zag flights. We then go on to ask what function these object-directed scanning movements might have in the context of landmark guided homing. Preliminary reports on this work have appeared elsewhere (Kelber and Zeil 1991; Voss and Zeil 1993).

Materials and methods

Wasps (*Odynerus spinipes*, Eumenidae) were studied at their nesting site in a small horizontal area of bare soil at the south-facing wall of Tübingen University's Chemistry building. The colony consists of about 50 individual nests which the females dig into the hard ground and which they provision with curculionid beetle larvae. Each nest entrance is marked by a small chimney which the resident wasp constructs with mud pellets. Chimneys are about 2 cm long and can rise vertically off the ground or may extend horizontally from the nest entrances. Their function is not clear: they may shelter the nest entrance from rain and parasites, but could also serve as a beacon to guide the wasp home. The insects are active from late May to early July.

A wasp can be induced to perform zig-zag flights by placing a novel object near the nest entrance while she is out foraging. The returning animal then performs extensive zig-zag flights for up to 20 s in front of the object. We either used a small metal cylinder (height: 7 cm, diameter: 1 cm) or a black styrofoam ball (diameter: 4.8 cm). When we positioned these objects we made sure that they would not obstruct direct access to a nest entrance. To study flight-control mechanisms, we attached a 3-cm styrofoam ball with a fishing line to the tripod, so that it either moved in the wind or could be set in motion by hand.

We recorded the behaviour of the wasps from above with a Hi-8 camcorder (Sony TR-705-E) which was fixed to a tripod 130 cm above ground. The recording area measured between 20 cm × 30 cm and 50 cm × 80 cm. We filmed about 70 sequences, lasting between 1.4 s and 12 s. Video-8 films were copied onto VHS tapes and analysed frame by frame. The sampling interval was 20 ms. The 3D flight paths and the orientation of insects were reconstructed automatically with the aid of a tracking program that has been described elsewhere (Voss and Zeil 1995). The software we developed estimates the height at which an animal flies by recording the position of its shadow on level ground. We could not resolve roll and pitch movements so that we have to neglect their possible effects. To reduce the noise in the data which is due to discrete sampling and to tracking errors, the flight paths were smoothed with a 1-8-1 filter for position and by a running average across three sampling intervals for the orientation of the insects. Scrutiny of high-speed films and of films recorded at large magnification showed that smoothing preserves the spatial and temporal aspects of the wasps' behaviour. At 50-Hz sampling rate, angular velocities and flight speed were determined over 40-ms intervals, for example:

$$\dot{\theta}_t = (\theta_{t+1} - \theta_{t-1}) \cdot 25.$$

We used the recorded flight paths and orientations of the wasps to reconstruct the optical flow they had experienced. The reconstruction was done by simulating a view from the 'cockpit' of the wasp into an artificial world, consisting of the ground plane, the horizon, the sky and one or more objects. Taking the relative position of wasp and object into account, we computed for each point of a regularly spaced grid on the ground and above the horizon, including the object edges, the changes in angular position caused by changes in position and orientation of the wasp flying along a zig-zag flight path. The resulting vectors were then projected frame-by-frame onto the computer monitor, using standard geometry. Our simulation can be seen as a 3D extension of a procedure developed by Lambin (1987), who reconstructed in the horizontal plane the field of image motion produced by walking crickets. The screen display does not represent the full, almost 360 degree panorama viewed by an insect's compound eyes. However, since the significant events in the case of zig-zag flights take place in the frontal visual field, we decided to neglect the rest of the field of view.

Results

Zig-zag flights and their control

Zig-zagging wasps (*Vespula vulgaris*) are a familiar nuisance at late-summer picnics, tea parties on the lawn or in beer gardens (see Atkinson 1993). The insects repeatedly approach objects like bottles, cakes and even faces and at close range start a series of rapidly swaying sideways movements, roughly perpendicular to their line of sight. In the wasp *O. spinipes* we discovered a rather similar flight behaviour directed towards novel objects in their usual flight corridor or in the vicinity of their nest entrances in the ground. During preliminary experiments with a variety of artificial objects we gained the impression that the intensity of these zig-zag flight manoeuvres and the readiness with which returning wasps would perform them, depend on special features of the introduced objects. Pieces of cardboard placed either flat on the ground or standing upright, do not seem to trigger zig-zag flights as effectively as 3D objects, like the black styrofoam ball and the small metal cylinder we eventually used.

Zig-zagging is most persistent when a wasp encounters an object for the first time. On consecutive returns to the nest from foraging excursions, the object-directed response decreases rapidly and after three or four encounters the wasp stops reacting to the novel object. Yet, when the object is repositioned or replaced by a different one, zig-zagging resumes. The occurrence, cessation and re-emergence of zig-zag flights is thus reminiscent of the 'turn-back-and-look' behaviour of honeybees at a newly discovered food place (Lehrer 1991, 1993).

A typical flight path of a zig-zagging wasp is shown in Fig. 1a. The wasp approaches the styrofoam ball in wide arcs from the lower left, zig-zags repeatedly at a distance of 10–15 cm away from the object and finally leaves the scene at the top of the figure. Throughout the sequence, the wasp is orientated roughly towards the object, while her direction of translation is almost perpendicular to her line of sight. Zig-zagging occurs most frequently at a distance of 7–11 cm from an object. Figure 1b demonstrates that the scanning movements are truly object directed: in this episode, the wasp turns away from her nest entrance (marked by a black dot) in the course of her swaying scrutiny of the styrofoam ball. When the object is set in motion when a wasp approaches, she adjusts both her orientation and her flight direction to keep the object in her frontal visual field and to maintain a flight path perpendicular to her line of sight to the object (Fig. 1c). Zig-zagging is thus under visual control. The time-course of various flight parameters is plotted in Fig. 2 for the sequence shown in Fig. 1a (see insets for definition of variables). The translational speed of the wasp oscillates uniformly at a frequency of 4–5 Hz, with maximum speed reaching 60–90 cm s^{-1} (Fig. 2a). Flight speed can drop to almost zero at the turning points. The orientation θ of the wasp's longitudinal body axis (Fig. 2b) is also modulated very regularly, with about half the frequency of flight speed, and almost in phase with her bearing relative to the object β (Fig. 2c). As a consequence of this close match between the time-course of θ and β, the wasp keeps the object in her frontal visual field throughout the recorded sequence. However, the image of the styrofoam ball is not held at a fixed retinal

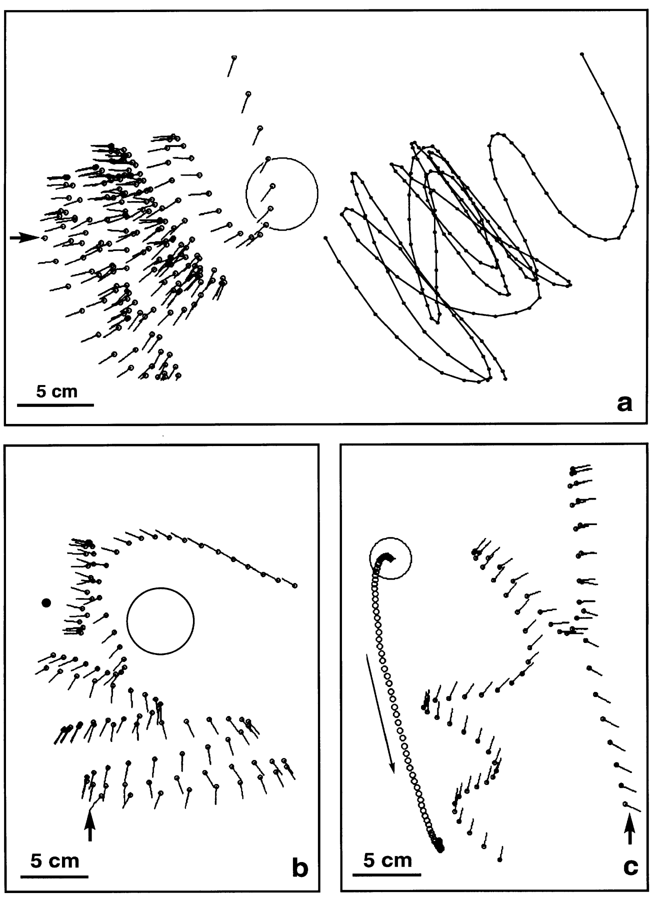


Fig. 1a–c The zig-zag flights of *Odynerus spinipes*. **a** A flight directed towards a black styrofoam ball (*large circle*) as seen from above. *Small circles with tails* mark the head position and the orientation of the longitudinal body axis of the wasp every 20 ms. The sequence is 3.7 s long, its beginning is marked by the *arrow*. The flight path on its own is shown on the right. **b** The flight path of a different wasp. The sequence is 2.16 s long. The *black dot* marks the nest entrance. **c** A zig-zag flight directed towards a moving object. The *large circle* marks the initial position of the object, the *smaller circles* the consecutive positions of its centre while it moved into the direction of the arrow. The sequence is 1.5 s long

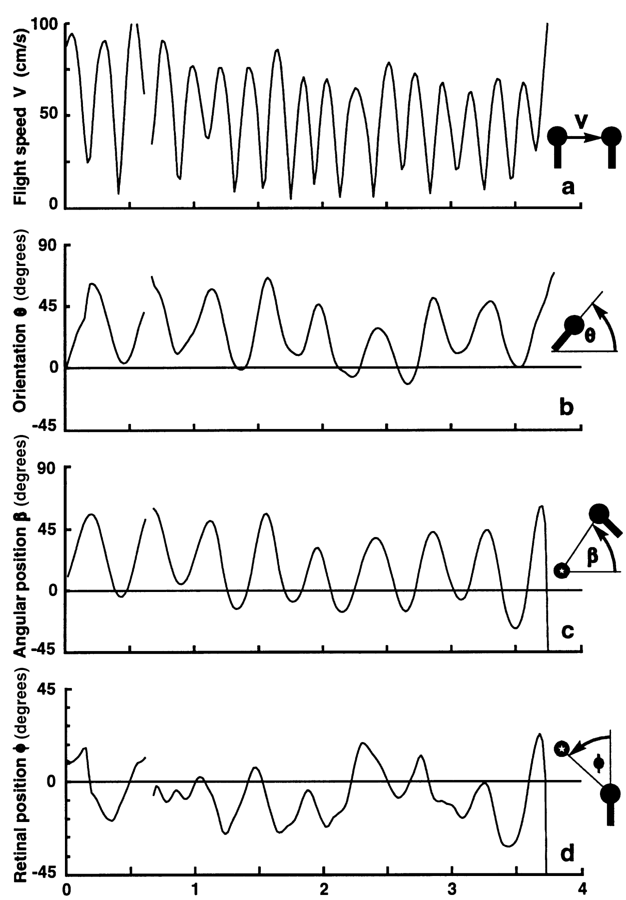
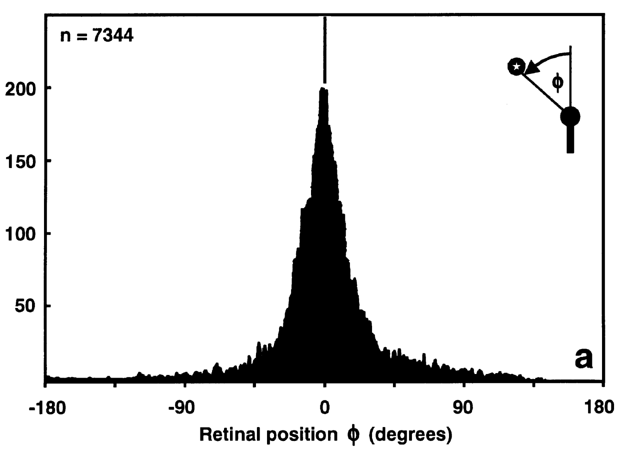


Fig. 2a–d The time-course of basic flight parameters for the sequence shown in Fig. 1a. See insets for definitions of variables. **a** Flight speed V (cm s^{-1}); **b** Orientation of the longitudinal body axis θ (*degrees*); **c** The wasp's bearing relative to the object β (*degrees*) which can be visualised as the orientation of the vector connecting the object's centre with the head of the wasp; **d** Retinal azimuth position of the object ϕ (*degrees*). Gaps in the record occur because sometimes the wasp's orientation cannot be determined

position but drifts to the right or to the left across the frontal visual field during each individual arc (Fig. 2d). We found no evidence, that compared to the centre of the styrofoam ball, other points on the ball or at its edges are fixated any more accurately. During the 70 zig-zag flights we recorded, the wasps viewed the centre of the styrofoam ball with frontal retina (Fig. 3a) and slightly below the horizon (Fig. 3b). The wasps thus always face their targets.

The retinal azimuth position of the object is controlled by the orientation of the insect's longitudinal body axis θ and of β, its bearing relative to the object (Fig. 4). We asked, how the wasps might control their turning velocity during zig-zag flights. A point to start with is to assume that the wasp's turning velocity $\dot{\theta}$ (Fig. 5a) is driven by the retinal position of the target φ (Fig. 2d). For 19 zig-zag flights, the best correlation we found was $r = 0.64 \pm 0.16$ (mean ± standard deviation, $n = 3294$) at a delay of 20 ms between samples of retinal position and turning velocity (see Fig. 8). Figure 6a, d show two individual examples. Maximal correlation between the retinal velocity of the object $\dot{\phi}$ (Fig. 5b) and the wasp's turning speed $\dot{\theta}$ (Fig. 5a) occurs at a delay of 100 ms with $r = 0.70 \pm 0.09$ (see Fig. 8 and individual examples in Fig. 6b,e).

In order to fixate an object while translating relative to it, a wasp would ideally need to match her turning speed $\dot{\theta}$ to her arc velocity $\dot{\beta}$ (Fig. 4). Judged by the correlation between these two parameters, this is exactly what the wasps do during zig-zag flights (Fig. 6c,f). For 19 flights we found an average correlation coefficient of $r = 0.87 \pm 0.05$ at 40-ms delay between the arc velocity and the turning velocity of wasps (see Fig. 8). Turning velocity thus seems to be driven by an input signal that is proportional to the arc velocity. The origin of this signal is not trivial: to measure her angular speed relative to an object, a wasp would need to know her distance to target, the direction of translation relative to it and her own flight speed. Wasps, however, could avoid these complications by using the difference between the retinal image velocity of the object and that of the background as an input signal to the system controlling turning velocity. The image motion $\dot{\phi}_R$ of all distant contours in the visual field of a wasp is determined exclusively by the wasp's own turning velocity $\dot{\theta}$, with $\dot{\phi}_R = -\dot{\theta}$ (see Fig. 4). If a wasp moved along a straight line without turning, the image motion $\dot{\phi}_T$ generated by a nearby object would in turn be exclusively determined by the



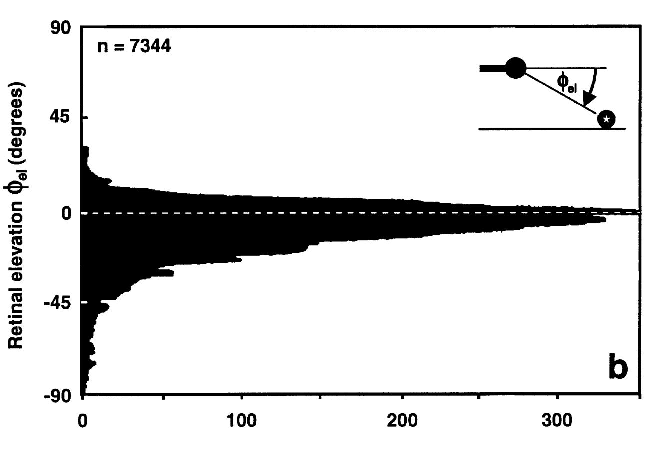


Fig. 3a, b The retinal position of the object during zig-zag flights. **a** Frequency distribution of retinal azimuth positions φ of the object centre during 70 zig-zag flights. Straight ahead at 0°, binwidth 1°. **b** Same for the retinal elevation ϕ_{el} of the object centre. The visual horizon is assumed to lie at 0°. *n*: number of samples taken at 20-ms frame intervals

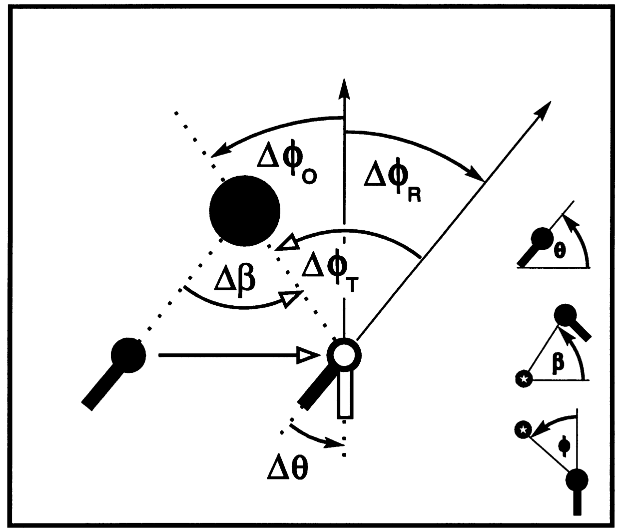


Fig. 4 The geometrical situation during object-directed zig-zag flights. When the wasp translates in one sampling interval to the right in front of an object without turning, the image of the object moves to the left in the visual field of the wasp by $\Delta\phi_T$, which is equal to the wasp's arc movement Δβ. The velocity of the object image depends on the speed and direction of translation and the distance of the object. When the insect then turns by Δθ, the object and all other contours in its visual field move to the right by $\Delta\phi_R$, which is equal to $-\Delta\theta$. When the wasp translates and turns at the same time, as is the case during zig-zag flights, the image of the object moves to the left by $\Delta\phi_O = \Delta\phi_T + \Delta\phi_R$. Neither $\Delta\phi_T$ nor Δβ can be directly measured by the wasp. To compensate for the apparent motion of the object owing to translation, the wasp must turn by $\Delta\phi_O$–$\Delta\phi_R$, i.e. the difference between object and background movement. Angular velocities are calculated by multiplying angular increments by the sampling rate

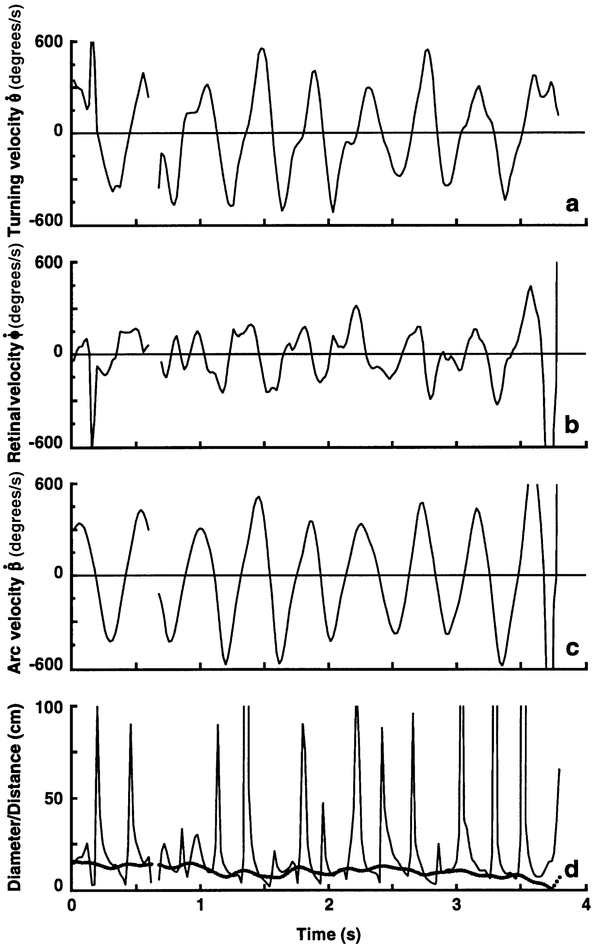


Fig. 5a–d The time-course of angular velocities and image motion for the same flight as the one characterized in Fig. 2. **a** Turning velocity $\dot\theta$ of the wasp (degrees s^{-1}); **b** Horizontal retinal velocity of the object $\dot\phi$ (degrees s^{-1}); **c** The wasp's arc velocity $\dot\beta$ (degrees s^{-1}) which is equivalent to the difference between the retinal velocity of the object $\dot\phi_O$ and the velocity of remote contours $\dot\phi_R$ (which is equal to $-\dot\theta$, see Fig. 4). **d** The *thin line* shows the diameter of the zero-horopter – the circle of zero horizontal image motion on the ground plane (cm) and the *thick line* the wasp's distance from the centre of the object (cm). See text for details

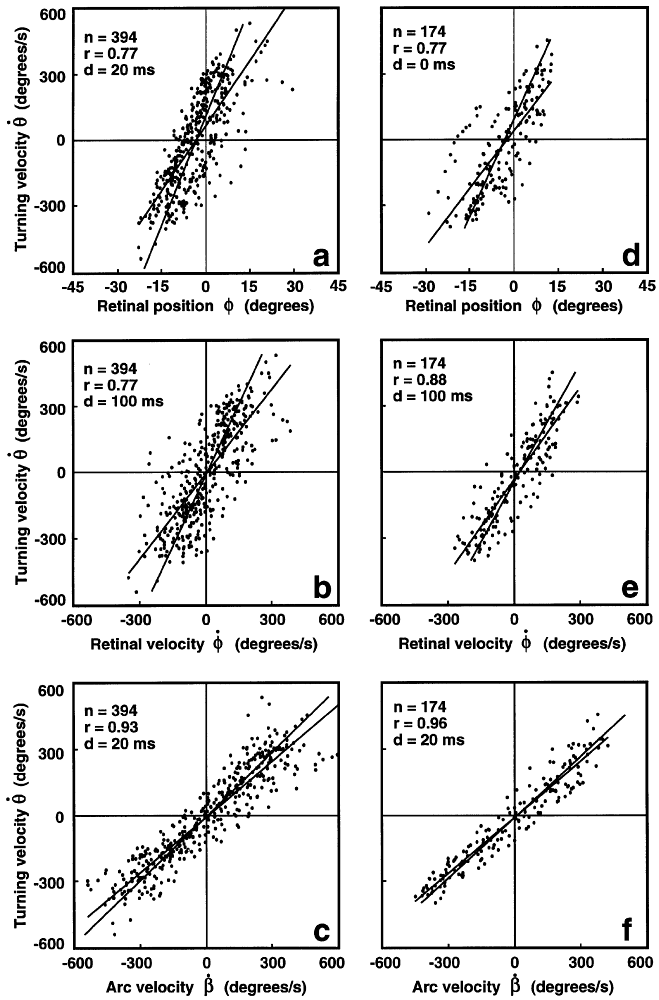


Fig. 6a–f Correlation analysis of two zig-zag flights (*left and right columns*). **a** and **d** Turning velocity $\dot\theta$ against retinal position ϕ of the object. Sample size *n*, correlation coefficient *r* and delay *d* are shown in the *top left corner* of the scatter plots. **b** and **e** Turning velocity $\dot\theta$ against the retinal velocity $\dot\phi$ of the object. **c** and **f** Turning velocity $\dot\theta$ against the arc velocity $\dot\beta$. The best correlation is found when turning velocity is shifted 20 ms backwards in time relative to arc velocity. This is the minimum delay we could measure at 20-ms sampling interval

insect's arc velocity $\dot\beta$. Now, since wasps both translate and turn during their zig-zag flights, the retinal velocity of the object is given by $\dot\phi_O = \dot\phi_T + \dot\phi_R$. To compensate for the effects of translation and hence to keep the object in the frontal visual field, $\dot\phi_R$ should be equal to $-\dot\phi_T$. The wasps could achieve this match by either minimizing $\dot\phi_O$ or by adjusting turning velocity so that $\dot\theta = \dot\phi_O - \dot\phi_R$, which represents the magnitude of the motion discontinuity at the object boundary when seen against a remote background. Note that $\dot\beta = \dot\phi_T = \dot\phi_O - \dot\phi_R$. $\dot\beta$ and $\dot\phi_T$ cannot be measured directly by the wasp, but the image velocities $\dot\phi_O$ and $\dot\phi_R$ can in a first approximation be determined from the activity of movement detectors. The unknown parameter $\dot\phi_T$ – the image motion component which is due to the wasp's translational movement – can thus be retrieved by subtracting the signals representing $\dot\phi_O$ and $\dot\phi_R$.

The time-course of turning velocities during zig-zag flights can indeed be described as a low-pass filter response to the motion discontinuity between object and background as input signal. Two examples are shown in Fig. 7 where the time-course of the low-pass filter output is plotted together with $\dot\theta$, the turning speed of the wasp and $\dot\phi_O - \dot\phi_R$, the input to the simulation. The optimal filter parameters vary between zig-zag flights and even within an individual flight. Low-pass filters of 2nd or 3rd order generally produce a good fit to the data.

A comparison of zig-zag and learning flights

Zig-zag flights share their pivoting structure with the learning flights performed by wasps and bees on de-

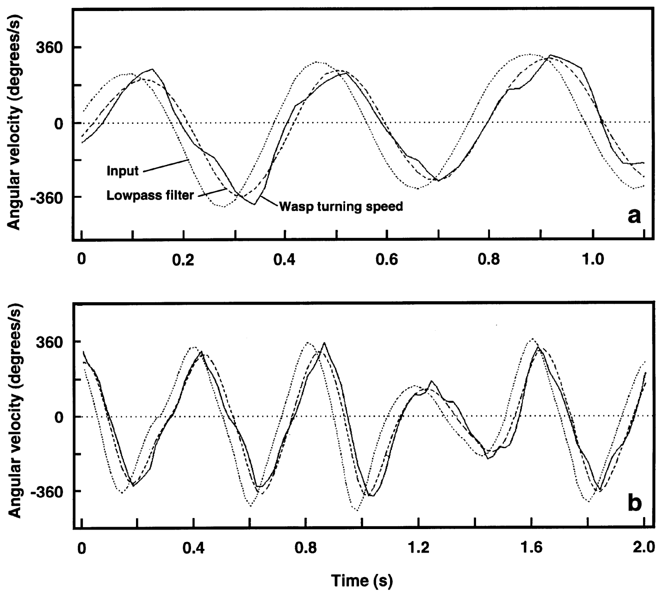


Fig. 7a, b A wasp's turning velocity $\dot{\theta}$ can be described by a low-pass filter response to $\dot{\phi}_O - \dot{\phi}_R$ as input signal (see Fig. 4 for definitions of variables). **a** and **b** show two examples taken from different zig-zag flights. The time-course of the turning velocty $\dot{\theta}$ of a wasp (*solid line*) is shown together with her arc velocity $\dot{\beta}$ (*dotted line*) which is equal to $\dot{\phi}_O - \dot{\phi}_R$, and the response of an optimized low-pass filter to $\dot{\phi}_O - \dot{\phi}_R$ as input (*dashed line*). **a** 2nd-order low-pass filter, $\tau = 0.0253$ s; **b** 3rd-order low-pass filter, $\tau = 0.01672$ s

parture from a goal (reviewed by Zeil et al. 1996). In contrast to zig-zag flights, however, learning flights are centred on an often inconspicuous goal which is fixated with lateral, not with frontal retina, as are the 3D objects during zig-zagging. We recorded a few learning flights of *Odynerus* and compared their structure and dynamics with zig-zagging flight paths.

Our results are summarised in Figs. 8 and 9. There are clear differences in the flight paths and in the time-course of orientation θ, bearing β and their derivatives (panels on the left in Fig. 8). The main reason for the differences is that a zig-zagging wasp reverses pivoting direction and direction of turning successively about two to three times a second while a wasp performing a learning flight keeps moving into one direction for up to 2 s at a time. During the learning flight, orientation and bearing are adjusted in such a way that the nest entrance is held at about 45° to the left and to the right of the wasp's midline, while the object during the zig-zag flight is viewed head on (see Fig. 8 top right histogram). The panels on the right of Fig. 8 show the results of cross-correlating the turning rate $\dot{\theta}$ with the retinal position of the pivoting centre ϕ (which is the nest entrance in the case of learning flights and the object in the case of zig-zag flights; top panels), with the retinal velocity of the pivoting centre $\dot{\phi}$ (centre panels), and with the arc velocity $\dot{\beta}$ (bottom panels) for 5 learning flights (left panels) and 19 zig-zag flights (right panels). Maximal correlation with a positive delay indicates that turning rate lags behind the other variables. The learning flights were all sampled at 25 Hz, so that the shortest delay we could measure was 40 ms. The correlation between turning rate and the other variables is generally weaker during learning flights and the maxima in the correlation functions are less pronounced compared to zig-zag flights. The apparent movement of the target during learning flights is caused by the wasp's rotational movements and not vice versa, as indicated by the fact that the best correlation between turning rate and the retinal velocity of the nest entrance is a negative one at 0 ms delay. In contrast, maximal correlation during zig-zag flights occurs with 100 ms delay. The cross-correlation analysis thus reveals a clear control structure of zig-zag flights, while the control variables of learning flights are more difficult to identify (see Zeil 1997).

Both flight manoeuvres share periodic changes in turning rate and arc velocity (Fig. 8) which, however, differ in frequency. The autocorrelation functions of these two variables document that the period length of 200–300 ms during the learning flights we analysed is shorter compared to the period length seen during zig-zag flights which is typically 400–450 ms (Fig. 9a, b). Zig-zagging in addition involves higher turning and pivoting velocities than learning flights (Fig. 9c, d).

The reconstruction of optical flow

The question that concerns us in this section is what the functional significance of the zig-zag manoeuvres might be. In contrast to jumping locusts, wasps do not show us by way of a distinct behaviour what they have learnt about a novel object. We therefore need to take an indirect route to approach this question. We tried to establish the visual consequences of these object-directed flight manoeuvres by mapping the topography of the image motion field they generate.

In a first step we determined the positions of minimal image motion every 20 ms during zig-zag flights which were performed in the presence of a moving object (as in the sequence shown in Fig. 1c). When a wasp moves in one sampling interval from position x_t, y_t to position x_{t+1}, y_{t+1} and changes her orientation from θ_t to θ_{t+1}, the horizontal image velocity in the visual field of the wasp is zero for objects that lie on a circle through x_t, y_t and

▶

Fig. 8 Comparison between learning flights and zig-zag flights. Flight paths are shown on top and the time-course of orientation θ, bearing β, and their derivatives for the same paths are shown on the lower left. The learning flight was sampled at 25 Hz, the zig-zag flight at 50 Hz. The *histogram at top right* shows the relative frequency of the retinal position of the nest entrance in the case of the learning flight (*dotted line*, $n = 107$) and the object in the case of the zig-zag flight (*grey area*, $n = 101$). Binwidth: 9°. *Panels on the right* show the results of a cross-correlation analysis between turning rate $\dot{\theta}$ and the variables retinal position ϕ, retinal velocity $\dot{\phi}$, and arc velocity $\dot{\beta}$, for 5 learning flights ($n = 578$) and 19 zig-zag flights ($n = 3294$). Positive delays indicate that changes in turning rate lag behind changes in the other variables. Means (*dotted thick line*) are shown plus/minus one standard deviation (*thin lines*)

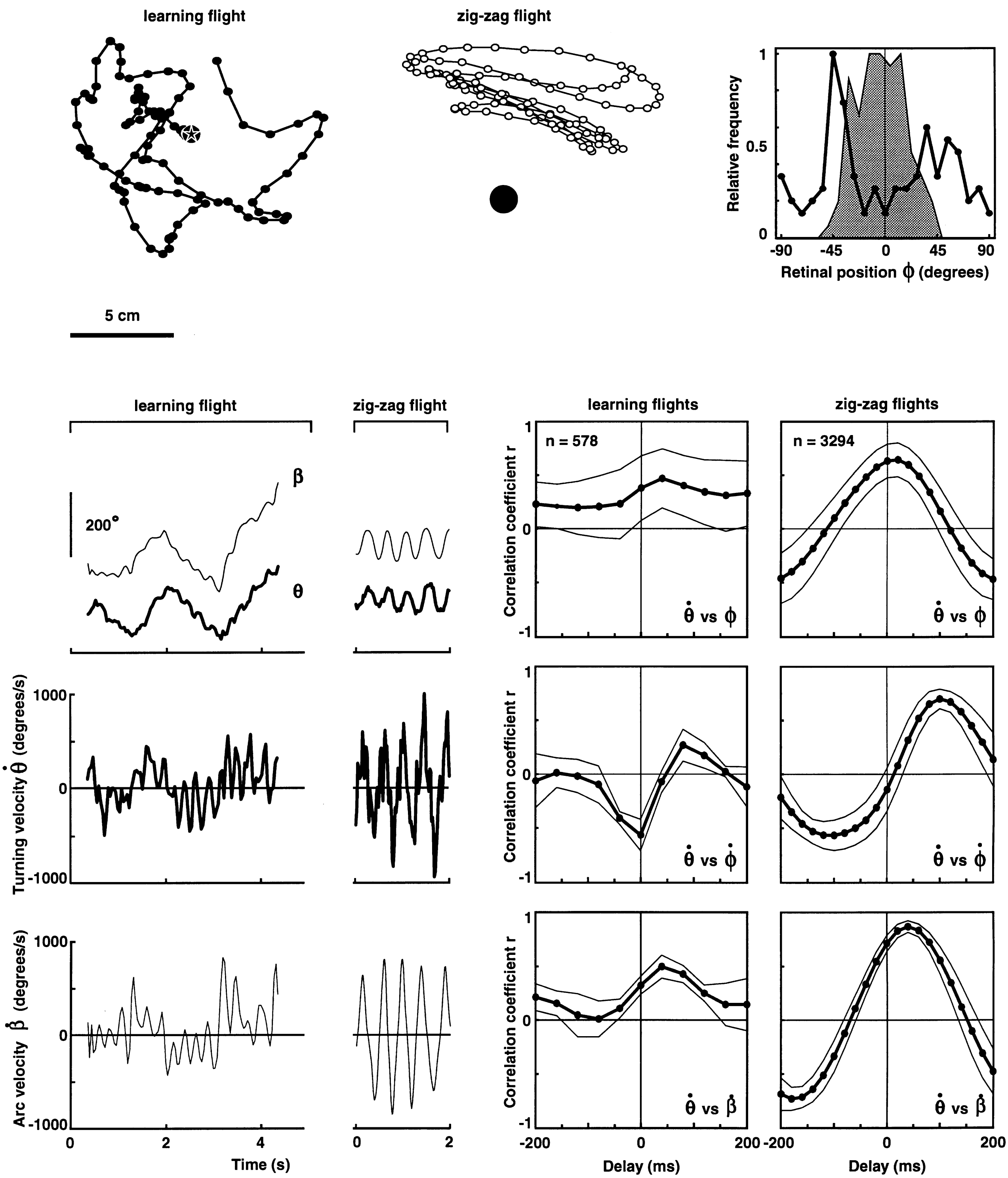
x_{t+1}, y_{t+1} (the zero-horopter). The radius r of the zero-horopter and the coordinates x_z, y_z of its centre are given by (Barth et al. 1991):

$$r = [\{(x_{t+1} - x_t)/2(\theta_{t+1} - \theta_t)\}^2 + \{(y_{t+1} - y_t)/2(\theta_{t+1} - \theta_t)\}^2]^{1/2};$$

$$x_z = (y_{t+1} - y_t)/2(\theta_{t+1} - \theta_t) + (x_{t+1} - x_t)/2; \quad \text{and}$$

$$y_z = (x_{t+1} - x_t)/2(\theta_{t+1} - \theta_t) + (y_{t+1} - y_t)/2.$$

The line running through the point half way between position x_t, y_t and x_{t+1}, y_{t+1} and the centre of the zero-horopter will intersect the horopter at distance $2r$ on the opposite side. We call this point of intersection the fixation point of the wasp. The scatter plot in Fig. 10 shows the fixation points for 36 zig-zag flights in a co-



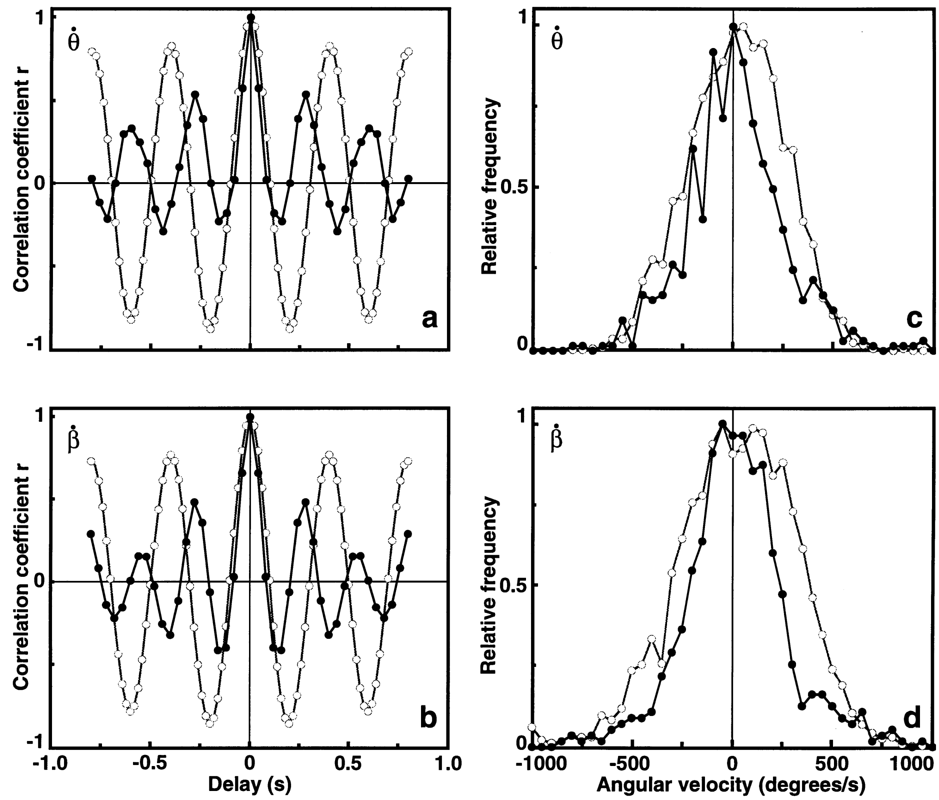


Fig. 9a–d Comparison between learning flights and zig-zag flights. **a** and **b** show the autocorrelation for turning rate $\dot{\theta}$ and arc velocity $\dot{\beta}$, respectively, for a segment of learning flight (*black dots*, *black line*) and a zig-zag flight (*open dots*, *grey line*). **c** and **d** Relative frequency of turning rate $\dot{\theta}$ and arc velocity $\dot{\beta}$ for 5 learning flights (*black dots*, *black line*; $n = 578$) and 19 zig-zag flights (*open dots*, *grey line*; $n = 3294$). Binwidth: $50° \text{ s}^{-1}$

ordinate system with the moving object at the origin. We included only instances in the analysis where the object speed exceeded 10 cm s^{-1}, to exclusively capture those segments of flight during which the wasps had to follow the object. The points form a cluster at the centre of the object, suggesting that the wasps' flight manoeuvres are indeed directed at the object and that it is the novel object itself which is the focus of attention.

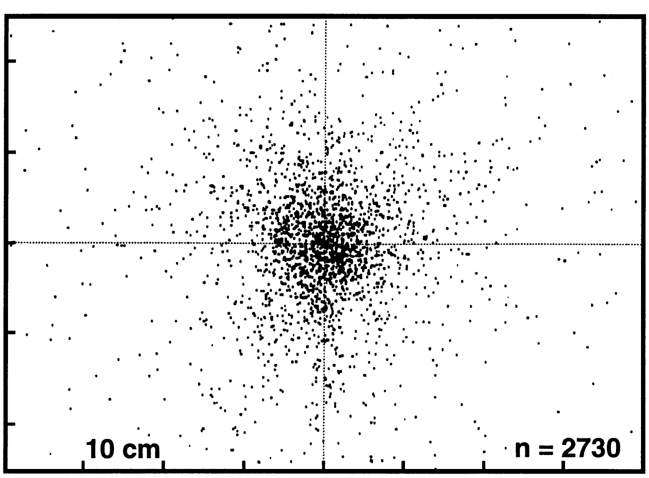


Fig. 10 The location of fixation points in the horizontal plane during 36 zig-zag flights which were directed at a moving object. See text for definitions and details. Coordinates of fixation points are expressed relative to the centre of the moving object at the origin of the plot. Although the object is moving, the fixation points are clustered around the centre of the object. Data were sampled from flight episodes where the object speed was larger than 10 cm s^{-1} ($n = 2730$)

We next attempted to analyse the dynamical aspects of the optical flow field by simulating a view from the 'cockpit' of a zig-zagging wasp. We are not yet in the position to reconstruct the real brightness distribution experienced by the insects and therefore created a simple and schematic world in which to realise this simulation. The insect is assumed to fly above plane ground that is stretching away to infinity where contours above the horizon are assumed to lie. The object at which the zig-zag flight is directed, however, is explicitly represented. Figure 11 shows two snapshots from the movie sequence for a zig-zag flight that was performed in front of a small metal cylinder. Each diagram represents an instant in the continuous display of the velocity vectors generated by distant contours above the horizon (large horizontal vectors in Fig. 11a) and regularly spaced contours on the ground. Figure 11a shows the motion vector field while the wasp is flying to the left and turning to the right. The fixation point during this segment of flight lies a small distance behind the object so that the object's image would move slowly to the right, while the distant contours above the horizon appear to move to the left at a velocity equal to the wasp's momentary rate of turning. This kind of shearing motion always occurs during the middle part of the insects' zig-zagging arcs. For a short moment at the beginning of arcs, the motion vector field becomes purely translational (Fig. 11b), with vectors on the ground plane all pointing away from the direction of heading and with distant contours being practically stationary on the wasp's retina.

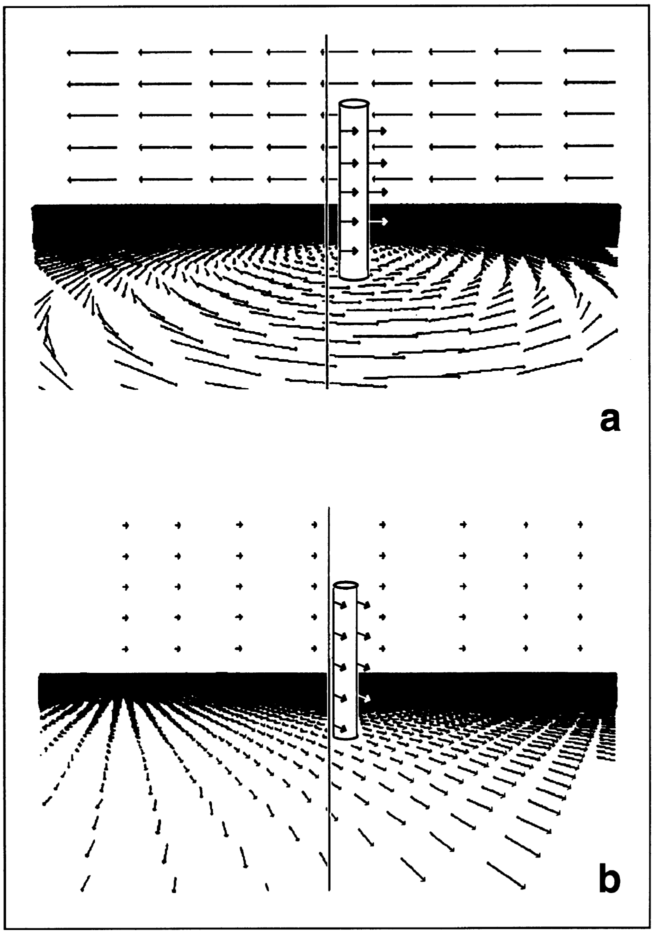


Fig. 11a, b Views from the 'cockpit' of a zig-zagging wasp. The simplified environment consists of the ground plane with regularly spaced dots extending to infinity where the contours above the horizon are assumed to lie. The field of view is about 60° wide and thus does not represent the full extent of the wasp's visual field. The object was a metal cylinder (7 cm high, 1 cm wide). **a** Shown is the field of motion vectors during the phase of a zig-zag flight where the ratio between rotation and translation is adjusted so that the object is near the centre of the vortex of motion vectors. **b** The field of motion vectors during an instant of nearly pure translation at the beginning of a new arc

The computer animations of typical zig-zag flights reveal some interesting facts beyond what we already know from the analysis presented earlier in this paper. As we have seen, the novel object remains for most of a zig-zag sequence within the frontal visual field, but oscillates across the midline. However, the dominant feature of the image motion field is the movement discontinuities between object and background. During the period of maximal angular and translational velocity, contours on the ground appear to rotate around the object, which then forms the centre of a vortex of velocity vectors (Fig. 11a). The object lies within or close to the centre of this vortex (the fixation point) for about 50% of the total flight-time. Each zig-zag cycle starts with a pure translation. The apparent movement of background contours is minimal during the initial phases of consecutive zig-zags, shortly after the wasps reverse their flight direction (Fig. 11b). All contours in the fronto-ventral visual field start moving in a direction opposite to the direction of translation, the close ones faster than the distant ones. Remote contours above the horizon do not move at all. Minimal image motion thus signifies remote contours. While a wasp's flight speed increases, she starts to turn and in her visual field the centre of the vortex moves closer and closer to the object and remains there for almost the second half of the cycle. Minimal image motion in this phase of the cycle thus defines the object of interest and other nearby contours. Image motion of background contours is equal and opposite to the direction of rotation of the wasp. Following this, a wasp's angular velocity decreases and a new translational phase into the opposite direction is initiated. The diameter of the zero-horopter, that is the horizontal distance between the wasp and the centre of the vortex field or the fixation point oscillates periodically throughout a zig-zag flight (Fig. 5d). Since each 'zig' and each 'zag' is initiated by a phase of pure translation, the diameter is large at the beginning of each cycle and then decreases exponentially with increasing rotational speed until it is close to the object distance, where in most cases it is held for the rest of the cycle.

It is not clear at present what triggers the end of an arc. The object itself may provide a relevant signal, for instance whenever its image passes the visual median. Alternatively, the turning points may be timed by an endogenous oscillator. Since the maximal values of translational and rotational velocities are fairly invariant throughout a zig-zag flight, it is also possible that a threshold for translational or rotational speed could provide a trigger. The wasps do not 'fixate' the scene at the endpoints of their zig-zags when translational speed is low as can be judged from the steep zero-crossings in Fig. 5a. In some cases, the turning points of consecutive zig-zags appear to lie along lines radiating from the object as in the orientation flights of *V. vulgaris* in front of a feeder (Collett and Lehrer 1993). However, for about 500 turning points in 36 zig-zag flights we could not find a significant tendency for systematic alignment. The computer animations confirmed this observation: alignment with background contours is not constant from one turning point to another. The instances of alignment we observed are rare and thus may occur accidentally.

Discussion

We have shown that the zig-zag flights of wasps are quite precisely controlled flight manoeuvres that have distinct consequences for the structure of the optical flow field experienced by the insects. The details of the image motion field suggest that it is specifically designed to acquire depth information. However, the wasps do not show any obvious change in their homing behaviour after they have performed zig-zag flights and it remains unclear to what extent the objects we introduced as a trigger of zig-zagging, are subsequently used as guiding landmarks by returning wasps. We thus cannot present clear evidence that these flight manoeuvres have a depth-scanning function. Yet there are a number of aspects concerning the context and the timing of zig-zag flights that should be considered before rejecting this possibility. Firstly, the flights are triggered by novel objects in an otherwise familiar environment. They therefore could be a way of inspecting changes in a scene and of up-dating the visual representation of it. Alternatively, they could be the result of instabilities arising when the insects try to match their stored representation to a scene that has changed. The flights in most cases are indeed performed by wasps returning home. Following the second hypothesis, it is hard to explain, however, why zig-zagging wasps should face the novel object and why they should track a moving one for some time before continuing their approach to the nest. Secondly, after a few encounters, the wasps stop directing zig-zag flights at a recently introduced object, a change in behaviour which is reminiscent of the learning dynamics of honeybees after they have discovered a new food source (Lehrer 1991, 1993). During their initial departures, the bees turn back, face the food source and perform characteristic learning flights. The active learning phase ends after the first eight or so visits, and the bees from then on do not turn back when they leave for their hive. Thirdly, the learning flights in bees and wasps and the object-directed zig-zag flights we describe here, share behavioural elements and the specific vortex field of image motion that the insects generate during their execution (Zeil 1993a). There is growing evidence that the prime function of learning flights in hymenopteran insects is the acquisition of depth information, most likely involving motion parallax cues (Zeil 1993b; Brünnert et al. 1994; Lehrer and Collett 1994; reviewed by Zeil et al. 1996).

The details of the optical flow generated during zig-zag flights thus may deserve a closer look to speculate on what information they might convey to the wasps. Our reconstruction of image motion revealed that for about half the flight time, the novel object is kept close to the centre of a vortex of velocity vectors. During this phase, image motion generated by the object is minimal, while background contours sweep through the visual field at a velocity equal and opposite to the wasps' turning velocity (Fig. 11a). Velocity contrast, therefore, is high at those retinal positions that view the edges of the object against a distant background; that is to say, at the upper parts of the object that extend above the wasps' height above ground. Velocity contrast will decrease towards the bottom part of the object, the edges of which are seen against the substrate. The centre of the vortex is in addition a region of large local differences in the direction and the relative size of velocity vectors. Conditions for the local detection of movement discontinuities – which provide information on object shape – are therefore optimal in the part of the visual field that views the centre of the vortex. Zig-zag flights generate both high-velocity contrast at object boundaries and good 3D resolution in the vicinity of the object itself. The insects thus produce what one may call a 'parallax fovea', specialized to retrieve shape from motion and the relation of an object to other features in the environment. Two observations suggest that zig-zagging indeed serves this purpose: the flights are preferentially performed in front of 3D objects and the wasps' turning velocity is driven by the velocity contrast at their edges. Beyond the velocity contrast signal, however, we do not know whether wasps are equipped with neural filters that can extract the local and global features of the pivoting parallax field during zig-zagging.

At this stage, the relationship between zig-zag flights and learning flights at the nest or at a food source is not clear. The structure and the dynamics of the two flight manoeuvres are clearly different in a number of respects, but we cannot offer a lucid interpretation. It may be that pivoting about a 3D object as opposed to an inconspicuous nest hole on the ground may require different flight control and image processing procedures. Beyond the role they may serve to scrutinise objects, the functional significance of zig-zag flights in the context of landmark guidance is equally hard to assess. After zig-zagging in front of a newly introduced object in the vicinity of their nests, wasps did not perform re-orientation flights on their subsequent departure as they normally do when they had problems in finding the nest during the previous return. The interest the wasps show in a new object decreases rapidly within the first three or four encounters and is reactivated only when the shape or the position of the object has changed. The flight manoeuvres thus seem to have an up-dating function that is called upon by a perceived mismatch between a stored representation and the actual scene and may serve to include novel features into the visual representation of the nest environment. Zig-zag flights differ from the re-orientation flights wasps and bees perform on departure from a goal area, in that they are directed towards changes in the vicinity of the goal during the approach and do not seem to be in any way related to the goal itself. The relationship between goal-directed and object-directed active learning will thus be an interesting topic for future research on landmark guidance.

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